

Video Processing and Optimisation

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July 15, 2026

1 Introduction

1.1 Problem Statement

Conventional methods for video compression are based on slicing frames and often lead to temporal discontinuity and poor results in video with motion. This research aims to develop a motion-compensated temporal frame fusion algorithm that reduces frames by integrating information from multiple consecutive frames onto one net frame, thereby preserving information and allowing data reduction.

2 Mathematical Background

Video could be thought of a 3D signal, with the domain being time ($\mathbb{R}$) and the co-domain being images ($\mathbb{R}$). At each instance of time, an image is being displayed, an image itself is a signal that maps a 2D space ($\mathbb{R}^2$) *onto light intensity of 3 colors (RGB)*, ($\mathbf{R} \times \mathbf{R} \times \mathbf{R}$)

3 Notes